



ROBOT LEARNING AND CONTROL GROUP MEETING REPORT

THESIS PROGRESS AND RESEARCH PLAN

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Outline

- ▶ Brief Self-Introduction
- ▶ Technical Pilot
- ▶ Thesis Project
- ▶ Future Work



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Brief Self-Introduction

1 Brief Self-Introduction

Background: B.Eng. in Computer Science and Technology, College of Computer Science, Chongqing University

Relevant Coursework: Advanced Mathematics, Linear Algebra, Probability and Statistics, Data Structures and Algorithms, Operating Systems, Machine Learning

Computational Strengths: Programming, algorithmic thinking, Linux-based development, machine learning practice, and end-to-end engineering implementation

Research Interest: Robot learning, dexterous manipulation, and sim-to-real transfer

Links: GitHub: github.com/CQULeaf Website: yexuhang.com



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Pilot Scope and Thesis Link

2 Technical Pilot

Purpose

- Learn the LEAP Hand robot-learning stack.
- Run a complete loop: modeling, training, playback, and ROS2 hardware deployment.
- Use the pilot to prepare for the thesis project, not to solve the same task.

Relation to Thesis

- Pilot: cube reorientation around the z axis.
- Thesis: continuous cylinder in-hand rotation.
- Shared pipeline: proprioception-only policy, relative joint targets, sim-to-real alignment.

Pilot task $\neq$ thesis task; pilot pipeline $\rightarrow$ thesis implementation baseline.



Task Modeling: From Goal to POMDP

2 Technical Pilot

Manipulation Goal

- Hold a cube inside the LEAP Hand.
- Track a sequence of local yaw goals.
- Reset or terminate when the cube leaves the hand or flips.

$$\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{O}, P, r, \gamma \rangle$$

$\mathcal{S}$ includes hand state, object pose, contact, and randomized dynamics; $\mathcal{O}$ is the deployable proprioceptive observation.

Goal Update

$$q_{k+1}^g = q_z \left(\frac{2\pi}{16} \right) \otimes q_k^g$$

$$d_R(q_t^o, q_t^g) = 2 \arcsin \left\| \text{vec}(q_t^o \otimes (q_t^g)^{-1}) \right\|_2$$

$$\text{success} = [d_R \leq 0.2] \wedge [\|p_t^o - p_{\text{hand}}\|_2 \leq 0.025]$$

$$\text{done} = [\|p_t^o - p_{\text{hand}}\|_2 \geq 0.07] \vee [\text{timeout}]$$



Observation, Action, and Reward

2 Technical Pilot

Policy Interface

$$\tilde{q}_t = \frac{2q_t - q_{\max} - q_{\min}}{q_{\max} - q_{\min}}$$

$$x_t = [\tilde{q}_t, u_t] \in \mathbb{R}^{32}$$

$$o_t = \text{vec}(x_{t-2}, x_{t-1}, x_t) \in \mathbb{R}^{96}$$

$$a_t \in [-1, 1]^{16}$$

$$u_t = \text{clip}\left(u_{t-1} + \frac{1}{24}a_t, q_{\min}, q_{\max}\right)$$

- No vision input.
- No tactile input.
- No object state as policy input.

Reward Design

$$r_t = w_p \|p_t^o - p_{\text{hand}}\|_2 + \frac{w_R}{d_R + \epsilon}$$

$$+ w_a \|a_t\|_2^2 + w_q \|u_t - q_{\text{grasp}}\|_2^2 + b_{\text{success}} + b_{\text{fall}}$$

- Position term keeps the cube near the hand.
- Rotation term tracks the current goal.
- Regularizers suppress large or unstable motions.
- Bonus/penalty handles success and falling.



Training in Isaac Lab

2 Technical Pilot

Environment

Isaac Lab DirectRLEnv
task registration
reset / reward / done

Sampling

8192 parallel envs
 $dt = 1/120$ s
decimation = 4

Policy

`r1_games` PPO
MLP [512, 256, 128]
GRU 256

Checkpoint

train / resume
sim playback
deployment restore

Isaac Lab rollout $\rightarrow$ PPO update $\rightarrow$ checkpoint $\rightarrow$ playback / deployment

Domain Randomization

- friction, mass, stiffness/damping
- initial pose noise and object wrench

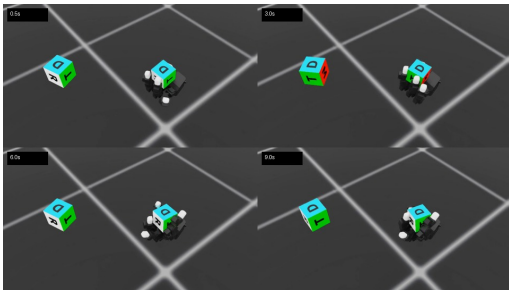
Control Robustness

- action noise and action latency
- no observation latency in this pilot



From Simulation to ROS2 Hardware

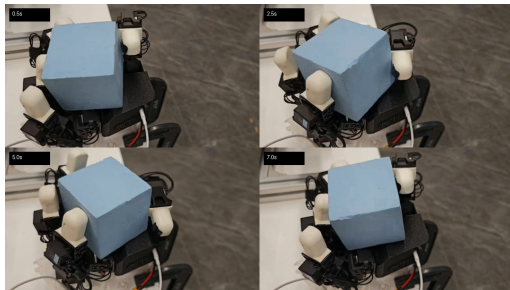
2 Technical Pilot



Simulation rollout

Checkpoint

restore trained policy



ROS2 hardware demo

Action

same relative target update

Observation

same 96D history

Key alignment issues: joint order, normalization, action scale, control rate, and safety limits.



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Thesis Project: Continuous In-Hand Rotation

3 Thesis Project

Research Question

- Can a low-cost LEAP Hand learn continuous in-hand cylinder rotation?
- Can the final policy run without vision, tactile sensing, or real object parameters?
- Where does the learned skill fail on real objects?

Pilot task → implementation baseline; thesis task → research contribution.

From Pilot to Thesis

- Pilot: discrete cube reorientation.
- Thesis: sustained cylinder rotation.
- Shared interface: proprioception and relative joint targets.



Task Definition: Sustained Rotation

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Goal

- Hold a cylinder inside the hand.
- Keep rotating around the vertical axis.
- Survive a fixed 20 s episode.

$$\begin{aligned}e_z &= [0, 0, 1]^\top \\ \Delta Q_t &= Q_t^o \otimes (Q_{t-1}^o)^{-1} \\ \omega_t^\parallel &= e_z^\top \frac{\text{Log}(\Delta Q_t)}{\Delta t}\end{aligned}$$

Key Difference from Reorientation

- No fixed terminal orientation.
- Reward measures continuous axial spin.
- Stability matters as much as rotation speed.

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]$$

Reorientation asks “where to stop”; rotation asks “how to keep going”.



Policy Interface: Deployable by Design

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Observation and Action

$$x_t = [\tilde{q}_t, q_t^{tar}] \in \mathbb{R}^{32}$$

$$O_t^\pi = [x_{t-2}, x_{t-1}, x_t] \in \mathbb{R}^{96}$$

$$h_t = [x_{t-29}, \dots, x_t] \in \mathbb{R}^{30 \times 32}$$

$$a_t \in [-1, 1]^{16}$$

$$q_t^{tar} = \text{clip} \left(q_{t-1}^{tar} + \frac{1}{24} a_t, q_{\min}, q_{\max} \right)$$

Training-Only Privileged Information

$$\xi_t = [p_t^o, \rho, m, \mu, c^\top]^\top \in \mathbb{R}^9$$

Signal	Real policy input?
joint position	yes
joint target history	yes
object pose	no
mass / friction / CoM	no
vision / tactile	no

The policy interface is constrained before training starts.



Reward: Rotate, Stabilize, and Control

3 Thesis Project

Rotation Term

$$r_t^{rot} = \lambda_{rot} \text{clip} \left(\omega_t^{\parallel}, r_{\min}, r_{\max} \right)$$

Total Reward

$$r_t = r_t^{rot} + r_t^{pose} + r_t^{lin} + r_t^{\tau} + r_t^{work} + r_t^a + r_t^{center}$$

$$\text{done} = [\|p_t^o - p_{\text{hand}}\|_2 \geq 0.07] \vee [t \geq T]$$

Design Intent

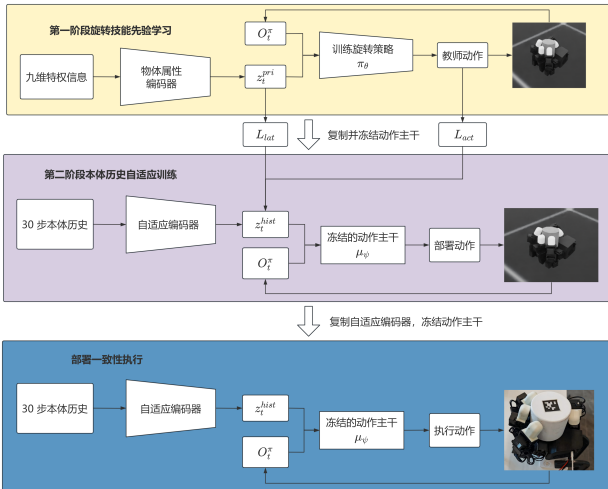
- reward axial spin, not arbitrary motion;
- keep the cylinder near the in-hand center;
- suppress linear velocity and pose drift;
- penalize torque, work, and action size;
- terminate on falling or timeout.

Reward design separates useful rotation from unstable throwing.



Skill-Prior-Driven Proprioceptive Rotation

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Stage 1

$$z_t^{pri} = E_{pri}(\xi_t)$$

Privileged PPO teacher.

Stage 2

$$z_t^{hist} = E_{hist}(h_t)$$

Train history encoder;
freeze actor.

Deployment

$$\mu_t = \mu_\psi([O_t^\pi; z_t^{hist}])$$

No privileged info.

Key idea

Train with privilege.
Deploy with
proprioception.



Training Protocol in Isaac Lab

3 Thesis Project

Simulation and Sampling

Item	Setting
Isaac Lab task	cylinder rotation
Physics / policy rate	120 Hz / 30 Hz
Decimation	4 physics steps
Episode length	20 s
Action scale	1/24 relative target
Evaluation	128 episodes per split

Inputs and Latent

$$O_t^\pi \in \mathbb{R}^{96} \quad h_t \in \mathbb{R}^{30 \times 32} \quad z_t \in \mathbb{R}^8$$

Actor input is short proprioceptive history plus the 8D skill prior.

Networks and Optimization

Module	Architecture	Used in
E_{pri}	MLP $9 \rightarrow 256 \rightarrow 128 \rightarrow 8$	Teacher
Actor/Critic	MLP [512, 256, 128]	Stage 1, deploy
E_{hist}	1D-CNN on 30×32 history	Stage 2, deploy

$$z_t^{pri} = \tanh(E_{pri}(\xi_t))$$

$$z_t^{hist} = \tanh(E_{hist}(h_t))$$

- Stage 1: PPO trains E_{pri} and actor/critic with 8192 envs.
- Stage 2: freeze actor, train E_{hist} with 4096 envs.
- Deploy-Refined: wider CoM range and $\lambda_{act} = 1$.

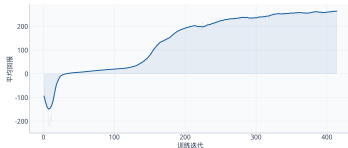
Deployment keeps only E_{hist} and the frozen actor.



Training Evidence: Successful Rotation

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Teacher Training

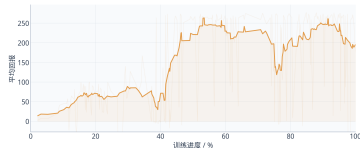


Teacher Playback

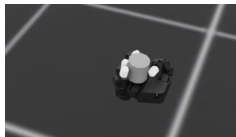


privileged policy learns stable axial spin

Deploy-Base Training



Deploy-Base Playback



proprioception-only policy can execute the skill



High-CoM OOD: Failure and Recovery

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Deploy-Base Failure



high CoM: contact drifts out

Base training solved nominal rotation, but high CoM shifts the contact patch and exposes a deployment-side stability gap.

Deploy-Refined Recovery



refined Stage 2 keeps contact

Observed Risk

$$|c| > 0.01 \text{ m} \Rightarrow \text{OOD contact drift}$$

Refinement

$$L_{stage2} = L_{lat} + \lambda_{act} L_{act}, \quad |c| \leq 0.015 \text{ m}$$



Simulation Results: ID and OOD Evaluation

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ID Evaluation

Policy	Success $\uparrow$	Survival $\uparrow$	Turns $\uparrow$	Spin $\uparrow$	LinVel $\downarrow$ cm/s	Torque $\downarrow$
Teacher	1.0000	0.9983	3.6950	1.1628	3.4478	2.3407
Deploy-Base	0.9922	0.9966	3.6309	1.1444	3.5378	2.3575
Deploy-Refined	1.0000	0.9983	3.7081	1.1669	3.6543	2.3373

OOD Evaluation

Policy	Success $\uparrow$	Survival $\uparrow$	Turns $\uparrow$	Spin $\uparrow$	LinVel $\downarrow$ cm/s	Torque $\downarrow$
Teacher	0.9766	0.9913	3.4684	1.0956	3.9620	2.4103
Deploy-Base	0.8750	0.9492	3.0422	0.9803	4.4935	2.4433
Deploy-Refined	0.9141	0.9758	3.2558	1.0420	4.2818	2.4087

Deploy-Refined improves OOD success, survival, net turns, spin, linear stability, and torque cost relative to Deploy-Base.



Ablations I: Deployable Skill Prior

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Privileged Information

Setting	ID success $\uparrow$	OOD success $\uparrow$	OOD turns $\uparrow$	OOD spin $\uparrow$
Full privileged information	0.9922	0.8750	3.0422	0.9803
no object position	0.9453	0.6172	2.3078	0.9231
no mass / friction	0.9375	0.7344	2.6264	0.9073
no CoM offset	0.0000	0.0000	0.4247	0.5707

Representation and Temporal Interface

Slice	Setting	ID success $\uparrow$	OOD success $\uparrow$	OOD spin $\uparrow$	Alignment $\downarrow$
representation	9D $\rightarrow$ 8D skill prior	0.9922	0.8750	0.9803	—
representation	direct 9D privilege	0.7734	0.5391	0.4005	—
history length	25 frames	1.0000	0.9219	1.0178	—
history length	30 frames	0.9922	0.8906	1.0500	—
encoder	Flatten-MLP	0.9688	0.7969	0.9036	0.4597
encoder	1D-CNN	0.9922	0.8906	1.0500	0.3018
encoder	GRU	1.0000	0.9141	1.0176	0.3147



Ablations II: Why Deploy-Refined Works

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Refinement Components under OOD

Stage 2 setting	Success $\uparrow$	Survival $\uparrow$	Turns $\uparrow$	Spin $\uparrow$	Action gap $\downarrow$
Deploy-Base	0.8750	0.9492	3.0422	0.9803	—
CoM only	0.8750	0.9469	3.0312	0.9847	0.3929
Action only	0.8906	0.9640	3.2001	1.0314	0.3635
CoM + action	0.9141	0.9758	3.2558	1.0420	0.3053

Action Consistency Weight

λ_{act}	ID S $\uparrow$	OOD S $\uparrow$	Survival $\uparrow$	Turns $\uparrow$	Spin $\uparrow$	Action gap $\downarrow$
0	0.9766	0.8750	0.9469	3.0312	0.9847	0.3929
0.25	1.0000	0.8828	0.9551	3.1595	1.0099	0.4317
0.5	0.9922	0.8906	0.9417	3.1117	1.0092	0.4594
1.0	1.0000	0.9141	0.9758	3.2558	1.0420	0.3053
2.0	1.0000	0.8672	0.9483	3.0868	1.0049	0.3466

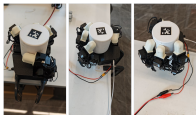
Action consistency gives the main gain; wider CoM provides the high-risk samples needed for the final combined policy.



Real Hardware: Transfer and Boundary

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Standard



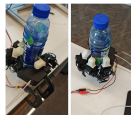
Smaller



Slender



Ribbed



Same Deploy-Refined Checkpoint, No Per-Object Retraining

Object	Physical change	Trials $\uparrow$	Est. turns $\uparrow$	Failure mode
standard cylinder	training-like cylinder	10/10	~ 0.5	none
smaller cylinder	smaller radius / lower mass	10/10	~ 1.0	none
slender bottle	near-cylinder, high aspect ratio	8/10	~ 1.0	tilt / fall
ribbed bottle	non-round local edges	0/10	—	edge jamming

Boundary: current policy transfers to cylinders and near-cylinders, but not ribbed non-round contact geometry.



Thesis Takeaways

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What Has Been Established

- A deployable LEAP Hand cylinder-rotation task.
- A two-stage skill-prior adaptation pipeline.
- ID/OOD simulation evaluation and ablations.
- Real hardware demos and failure boundary.

Current claim: cylinder and near-cylinder rotation, not arbitrary-object rotation.

Open Problems

- Real rotation speed is much lower than simulation.
- Real object rotation is manually estimated.
- Non-round local geometry still breaks contact progression.
- Broader object generalization needs better sensing or richer training coverage.



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From Joint-Space Policy to Hand Geometry

4 Future Work

Current Policy

- 16D relative joint targets.
- Proprioception-only deployment.
- Cylinder and near-cylinder rotation works.

Observed Boundary

- Ribbed or non-round geometry causes contact jamming.
- Hand shape and contact geometry are implicit.
- Generalization depends heavily on training coverage.

Research Problem

Can contact-relevant hand keypoints reduce learning complexity while preserving dexterous adaptability?

Structure: contact-envelope action prior **Freedom:** learned residual correction



Affine Transformation on Hand Keypoints

4 Future Work

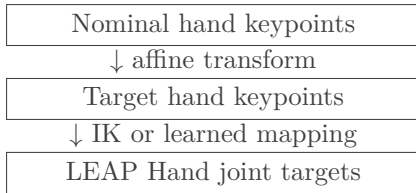
Key Representation

$$P_i^* = AP_i^0 + b$$

- P_i^0 : nominal contact-relevant hand keypoint.
- A : scaling, rotation, and shear.
- b : translation.
- P_i^* : target hand-shape keypoint.

The affine transform is applied to hand keypoints, not directly to joint angles.

Control Pipeline



Candidate nodes: palm reference, finger-link surface points, fingertips, and contact proxies.



Initial Research Plan

4 Future Work

1. Define Contact Keypoints

- palm / in-hand reference;
- finger-link surface points;
- fingertips only if useful.

2. Build Affine Prior

- start from nominal grasp shapes;
- scale, rotate, and shear keypoints;
- map keypoints to joint targets.

3. Compare Interfaces

- direct 16D joint-space policy;
- affine-only policy;
- affine plus residual policy.

Evaluation

- object size variation;
- object orientation variation;
- near-cylinder versus non-round geometry.

Goal: keep the existing sim-to-real pipeline, but improve geometric generalization.



Q&A

Thank you for listening!
Your feedback will be highly appreciated!